

KAZI MD ZAREER

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Portfolio ♦ ResearchGate ♦ LinkedIn

EDUCATION

Bachelor of Science in Mechanical Engineering

Jan. 2022 - Jun. 2026

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Cumulative GPA: 3.57/4 (Major GPA: 3.91)

Relevant Coursework: Mechanics of Solids | Mechanics of Machinery | Machine Design | Applied Engineering Mathematics | Composite Materials | Control Engineering | Heat Transfer | Fluid Mechanics | Instrumentation and Measurement | Bio-engineering

RESEARCH INTERESTS

Additive Manufacturing · Finite Element Analysis · Computational Mechanics · Computational Heat Transfer · Machine Learning

RESEARCH EXPERIENCE

Undergraduate Thesis, Department of Mechanical Engineering, BUET

Jun. 2025 - Jun. 2026

- Employed finite element-based computational methods to investigate how variations in key LPBF process parameters influence the mechanical integrity of fabricated components, with emphasis on stress concentration and stress intensity factor-related behavior

Computation Fluid Dynamics and Heat Transfer (CFDHT) Research Group, Department of Mechanical Engineering, BUET

Sep. 2024 – May 2025

- Magnetohydrodynamics (MHD) Mixed Convection and Entropy Generation Analysis of Cu–Water Nanofluid in a U-Shaped Lid-Driven Cavity with a Conjugate Elliptical Obstacle and Corner Heating
- Performed finite element-based simulations in COMSOL Multiphysics; completed model validation against benchmark lid-driven cavity flow and MHD entropy generation studies

PUBLICATIONS

Journal Papers

[1] **K. M. Zareer**, R. Mostakim, M. A. Motalab, “Residual Stress – Fracture Mechanics Correlation in Laser Powder Bed Fusion of 17-4PH Stainless Steel: A Parametric Investigation of Stress Concentration and Stress Intensity Factors,” [Under Review]

Conference Papers

[1] M. R. Rifad, **K. M. Zareer**, M. A. O. R. Hossain, M. M. H. Ashik, “GreenGuardian: A Quadruped Spider Robot Prototype for Real-Time Potato Disease Detection Using Deep Learning,” *International Conference on Sustainable Agriculture and Smart Development (ICSASD)* [In Press — Best Paper Award]

PROJECTS

GreenGuardian: Quadruped Spider Robot for Potato Disease Detection

Nov. 2023 – Mar. 2024

- Built a 3D-printed quadruped robot (12 DOF) with ESP32-CAM live video and a custom CNN classifying Early Blight, Late Blight, and Healthy leaves at 95% accuracy
- Designed an end-to-end sensing-to-prediction pipeline for terrain-adaptive crop monitoring at low hardware cost

Comparison of Algorithms for Tuning PID Controllers Used in a Two-Wheeled Self-Balancing Robot Apr. 2025 – Jun. 2025

- P, PI, and PID controllers tuned by Ziegler–Nichols, MATLAB auto-tuner, Genetic Algorithm (GA), and Clonal Selection Algorithm; GA-tuned PID achieved zero overshoot and 0.24% undershoot, outperforming all other strategies

Cooling Load Calculation and Air Conditioner Sizing for a Residential Room May 2025 – Jun. 2025
DOI: 10.13140/RG.2.2.13128.89608

- Calculated total cooling load (≈ 0.6 TR) of a residential room per ASHRAE CLTD/CLF methodology, accounting for wall/roof/door/glass conduction, solar gain, infiltration, and internal loads

Design and Fabrication of a STHX with Disk-and-Doughnut Baffles Aug. 2024 – Dec. 2024
DOI: 10.13140/RG.2.2.32077.60641

- Designed, simulated, and fabricated a shell-and-tube heat exchanger (STHX) with disk-and-doughnut baffles; tested up to 300 psi with no leakage and compared baffled and unbaffled configurations, demonstrating improved heat transfer with alternating disk-and-doughnut baffles.

Stress Analysis of Key Components in a Crane Pulley System: A Combined Finite Element and Analytical Study Oct. 2024 – Dec. 2024
DOI: 10.13140/RG.2.2.23037.93928

- Performed FEA-based stress/strain/deformation analysis of a crane hook, V-groove pulley, and cantilever beam in ANSYS; validated hook stress against hand-calculated curved-beam (Winkler-Bach) theory

INDUSTRY EXPERIENCE

United Ashuganj Energy Ltd., 200 MW IPP Power Plant 10 - 25 Mar. 2025
Engineer Intern — DOI: 10.13140/RG.2.2.19525.20967 *Ashuganj, Bangladesh*

- Trained in scheduled maintenance, troubleshooting, and performance assessment of plant equipment across the workshop, boiler-turbine section, and engine hall, gaining a working understanding of combined-cycle power generation, SCADA-based control, and grid integration

TECHNICAL SKILLS

Programming	Python, MATLAB, C/C++, Arduino, HTML/CSS, Tecplot $\LaTeX$
CAD & Simulation	ANSYS, COMSOL Multiphysics, SolidWorks, AutoCAD, Tinkercad, Fritzing

AWARDS AND ACHIEVEMENTS

- ME faculty **Dean’s List Award**, **University Merit Scholarship** from BUET
- 6th Board, **Best Player**, BUET Inter-Hall Chess Tournament 2024
- **2nd Runner-Up**, Bangladesh Mathematical Olympiad (BdMO), National Round 2018
- **Board Scholarships**, e.g., HSC, SSC, JSC, PECE

CERTIFICATIONS

- **COMSOL Multiphysics Simulation of Thermofluidic Problems (Advanced Level)** Nov. 2024
Directorate of Continuing Education (DCE), BUET
- **Machine Learning Specialization** Sep. 2025
Stanford Online | DeepLearning.AI | Coursera